KOSZUL-REGULARITY OF AN IMMERSION NEAR A POINT DOES NOT DEPEND ON THE CHOSEN SMOOTH FACTORIZATION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 18072-18081 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 18083-18091, the complete author proof (7 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Well-definedness lemma feeding the chapter's lci/perfect formalism (used by the local complete intersection morphism independence arguments).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Divisors, lemmas lift-regular-immersion-to-smooth and push-regular-immersion-thru-smooth (the two transport laws for Koszul-regular immersions through smooth morphisms).

Source reference. Original lemma 067S: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18091–18091. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0693: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18091–18091. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let U, P, P' be schemes over S. Let $u \in U$. Let $i : U \to P$, $i' : U \to P'$ be immersions over S. Assume P and P' smooth over S. Then the following are equivalent*

(1) *i is a Koszul-regular immersion in a neighbourhood of u, and*
(2) *i' is a Koszul-regular immersion in a neighbourhood of u.*

Proof. Assume i is a Koszul-regular immersion in a neighbourhood of u. Consider the morphism $j = (i, i') : U \to P \times_S P' = P''$. Since $P'' = P \times_S P' \to P$ is smooth, it follows from Divisors, Lemma 067S that j is a Koszul-regular immersion, whereupon it follows from Divisors, Lemma 0693 that i' is a Koszul-regular immersion. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.